Language -> communication tool.
Program -> set of instruction.

used to communicate with computers to perform set of instruction.

Types

Low Level

1) Assembly Language
2) Machine Language. (0-1)

high Level.

1) Procedural Oriented.
C

2) Object oriented
Python

3) function oriented. (JavaScript)

Compiler/Interpreter

Java is Partially Object oriented.

Compiler -> Translate high level Programming Language into Machine Language. <- Interpreter

Compiler	Interpreter
* Translates the entire Program.	* Translates Line by Line.
* Time Take is Less	* Time Taken is more.
* Difficult to debug	* Easy to debug.
* C, C++	* Interpreter.

Java

-> it was developed by James Gosling at Sun micro Systems.

Structure of java

Documentation Section

Single line Comment

//

Multi Line Comment.

/* ——— */

Class initialization — class Main {
}

file name

main() — Public Static void main(String args[]) {
}

Static method will access only Static variable / method

output statement

-> System.out.Print

-> System.out.Println()

1st Print then if it is a 'ln' it will create a new line.

```
System.out.print("Nanthish");
System.out.print("Samantha");
System.out.println("Nanthish");
System.out.print("Samantha");
System.out.println("Nanthish");
System.out.print("Samantha");
```

Escape sequence

\n - new Line

\t - tab space.

Number System

binary(2) — 0-1
octal(8) — 0-7
decimal(10) — 0-9
hexa decimal(16) — 0-9, 10-A, 11-B

Decimal to any base

2	33
2	16 - 1
2	8 - 0
2	4 - 0
2	2 - 0
2	1 - 0
	0 - 1

⇒ (1 0 0 0 0 1)

2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1
1	0	0	0	0	1

10 ⇒ 1 0 1 0
20 ⇒ 1 0 1 0 0

32	16	8	4	2	1	
1	0	1	0	1	0	⇒ 42

---

6 bits representation

(MSB) 2^5	2^4	2^3	2^2	2^1	2^0 (LSB)
-32	16	8	4	2	1
_	_	_	_	_	_

neg representation.

$[-32, 31]$
$[-2^5, 2^5-1]$

4 bits

MSB 2^3	2^2	2^1	2^0
8	4	2	1
_	_	_	_

$[-8, 7]$
$[-2^3, 2^3-1]$

Data Types

collection of raw facts

byte — 1 byte = 8 bits $[-2^7, 2^7-1]$ => $[-128, 127]$

short - 2 byte = 16 bits $[-2^{15}, 2^{15}-1]$

int - 4 bytes = 32 bits $[-2^{31}, 2^{31}-1]$

long - 8 bytes = 64 bits $[-2^{63}, 2^{63}-1]$

— whole Number.

Point Values

float - 4 bytes — 32 bits $[-2^{31}, 2^{31}-1]$

double - 8 bytes — 64 bits $[-2^{63}, 2^{63}-1]$

char - 2 bytes - 16 bits $[-2^{15}, 2^{15}-1]$

boolean - 1 bit [true / false]

Primitive data type	Non-Primitive data type
byte, short, int, long, float, double, char, boolean	Array, ArrayList

Variable

used to store the address of data.

Syntax

data_type var_name;

int num;

num = 22;

num → #120

22

4 bytes.

Rules of Variables (S, S.D, K)

1) Space is not allowed.
2) Special symbol is not allowed Except (_, $)
3) Should not start with Digits.
4) Variable name can not be a keyword

Reserved words.

Value initializing to Variable.

1) Compile Time / static initialization

```
public class Main{
    public static void main(String args[]){
        int num=22;
        System.out.println(num);
    }
}
```

2) Run Time / dynamic Initialization (getting Value from user).

1) import Scanner Class

2) Scanner obj_name = new Scanner(System.in);

you can get the value from user.

int

```
Scanner scn=new Scanner(System.in);
int num=scn.nextInt();
System.out.println(num);
```

```java
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        //String val=scn.next(); it reads one word
        //String val=scn.nextLine();
        //int num=scn.nextInt();
        //float val=scn.nextFloat();
        //double val=scn.nextDouble();
        //char val=scn.next().charAt(0);
        System.out.println(val);
    }
}
```

Operators :- it is a symbol which is used to perform operation between any 2 operands.

a=10, b=5

$\boxed{c=a+b}$

1) Arithmatic operator (+, -, *, /, %)

10/5 = 2	10%5 = 0	7/9
8/3 = 2	8%3 = 2	7%9
10/2 = 5	10%2 = 0	

5 = /

2 | 10

10

0 = %

UV < LV

/ → 0

% → UV

9/7 = 1	7/9 = 0
9%7 = 2	7%9 = 7

Functions → set of instruction to perform specific task.

Function

Pre defined — max(), sqrt()

User defined

1) with argument & with return type.

```java
import java.util.*;
public class Main{
    public static int sum(int num1,int num2){
        return num1+num2;
    }
    public static void main(String args[]){
       Scanner scn=new Scanner(System.in);
       int a=scn.nextInt();
       int b=scn.nextInt();
       int res=sum(a,b);//function call
       System.out.println(res);
    }
}
```

2) with arguments & without return type.

```java
import java.util.*;
public class Main{
    public static void sum(int num1,int num2){
        System.out.println(num1+num2);
    }
    public static void main(String args[]){
       Scanner scn=new Scanner(System.in);
       int a=scn.nextInt();
       int b=scn.nextInt();
       sum(a,b);
    }
}
```

---

2) Assignment operator (=)

a = 10;

a = a+5; — a+=5;

a = 5

a = a*5; — a*=5;

---

3) Relational operator (>, ≥, <, ≤, ==, !=)

9 > 2 — True

2 > 9 — False

9 < 20 — True

20 < 9 — False

5 == 4 — F

5 != 4 — True.

4) Logical Operator (&&, ||, !)

		&&	||	NOT
(5>3) T	T	T	T	T -> F
T	F	F	T	F -> T
F	T	F	T	
F	F (3>5)	F	F	

5) Unary Operator (Increment / Decrement operator)

Increment: ++

Decrement: --

a=5

Pre Increment: ++a

Same Line a=a+1

Post Increment: a++

Same Line a= old value

a=a+1

6) Condition / Ternary Operator (?:)

(Condition)?(True statement):(false statement);

```
Scanner scn=new Scanner(System.in);
int n1=scn.nextInt();
int n2=scn.nextInt();
int max=(n1>n2)?n1:n2;
System.out.println(max);
```

```
class Solution {
    public int greatestOfThree(int a, int b, int c) {
        // code here
        return (a>b)?(a>c?a:c):(b>c?b:c);
    }
}
```

greatest of 3 number

Bitwise Operator (&, |, ^, <<, >>)

		&	|	^ (Same Same Puppy Same 0)
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

5 — 101

7 — 111

& 101 — 5
| 111 — 7
^ 010 — 2

① odd or Even (without %)

...., 2^5 2^4 2^3 2^2 2^1 2^0 — odd

32 16 8 4 2 1

n ⇒ 1 0 0 1 0 1 | 1 0 0 1 0 0

(&) 1 ⇒ 0 0 0 0 0 1 | 0 0 0 0 0 1

0 0 0 0 0 1 | 0 0 0 0 0 0

(n & 1) — 1 (odd)
(n & 1) — 0 (Even)

n % 2 == 0 — Even
== 1 — odd

(n & 1) == 0 — Even,
== 1 — odd.

Left-Shift

n		MSB				LSB	
1	=>	0	0	0	0	1	— (1<<1)
2	=>	0	0	0	1	0	— (2<<1)
4	=>	0	0	1	0	0	— 4<<1
8	=>	0	1	0	0	0	

$n<<1 = n*2$

$n<<1(2^1) => n*2$

$n<<2(2^2) = n*4$

$n<<3(2^3) = n*8$

Right shift

		32	16	8	4	2	1	
10	—	0	0	1	0	1	0	(10>>1)
5	—	0	0	0	1	0	1	(5>>1)
2	—	0	0	0	0	1	0	(2>>1)
1	—	0	0	0	0	0	1	

$n>>1 = n/2$

$n>>2 = n/4$

$n>>3 = n/8$

```java
//Age Eligible
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        int age=scn.nextInt();
        System.out.println(age>21?"Eligible":"Not Eligible");
    }
}
```